

ARYAN MADHUR HAMINE

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UC Davis CS graduate student and software engineer with 2.5 years shipping production healthcare-tech SaaS in Python/Django, Node.js, and React, backed by PostgreSQL/MySQL, Redis, Elasticsearch, and AWS. Built agent memory, notification, and evaluation systems for SAP's Joule assistant, and researching real-time audio and observability for agentic AI systems at UC Davis.

EDUCATION

University of California, Davis, USA

Sept 2025 – Jun 2027 (Expected)

Master of Science in Computer Science (GPA 3.85/4)

Key Coursework: Distributed Database Systems, Machine Learning and Discovery

MMCOE, Pune University (SPPU), India

Aug 2019 – Apr 2023

Bachelor of Engineering in Information Technology (GPA 3.7/4)

Key Coursework: Data Structures and Algorithms, Operating Systems, Cloud Computing, Software Engineering

PROFESSIONAL EXPERIENCE

SAP - Palo Alto, CA, USA

Jul 2026 – Sept 2026

AI Software Developer Intern

- Architected an autonomous **LangGraph batch notification pipeline** on the Joule learning assistant that processes learner data uploads through a two-tier **orchestration graph**, handles each learner in parallel, and reliably delivers notifications through a retry-backed sender with **exponential backoff**
- Designed a reusable memory framework on **HANA Cloud** that gives the AI agent persistent, tenant-isolated storage with **content-hash keying** and **freshness control**, and exposed it through **LangChain** tools for cross-conversation recall of user preferences
- Raised the agent's automated test pass rate from **16% to 94%** by fixing cases blocking the CI/CD pipeline, benchmarked frontier models, and **hardened safety guardrails** against out-of-scope responses by reworking the **system prompt**

CreliaHealth - Pune, India

Apr 2023 – Aug 2025

Software Engineer (SDE-II)

- Engineered **Storage Manager**, a unified Django REST service handling **1M+ uploads** with Elasticsearch indexing and MySQL-based lab tracking; achieved **~40% faster throughput** and complete upload traceability
- Enabled **10k+ patients/month** to access medical imaging and reduced support tickets **30%** by shipping a secure **PACS/DICOM sharing portal** with authenticated links and audit logs
- Engineered **observability into webhook integrations**, increasing reliability by **95%** and saving **20+ engineer-hours weekly**, by replacing Mirth-based **retry workflows** with a Django-DocumentDB integration dashboard and an auto-retry mechanism that re-executed failed webhooks using recorded state data and **AWS EventBridge scheduling**
- Reduced billing errors** and eliminated manual steps by automating reflex-testing rules over MySQL and **caching thresholds in Redis** for real-time evaluation

PROJECTS

Aura - Policy-Driven Selective Observability for Agentic AI Systems

- Built a stateful policy engine that escalates telemetry detail at runtime on agent-semantic failure signals, gating field-level payload capture instead of flat traces; in a **10,000-event evaluation**, retained **~96% diagnosability** while cutting telemetry volume **~34%**
- Designed a travel-agent harness using LangGraph and LangChain (**200 runs, 9 failure scenarios**) with ChromaDB memory, multi-tool orchestration, and ICS calendar export to generate realistic agent traces
- Built a **deterministic per-run evaluator** computing rubric scores, hallucination ratios, and PII detection over the agent harness

Cochlearity - Gaze-Steered AR Audio Pipeline (Miller Lab, UC Davis, Jan 2026 – Present)

- Building the **real-time** end-to-end pipeline for an **assistive AR hearing system** combining a multichannel mic array, **gaze-driven steering**, and on-device speech recognition
- Integrating eye-tracker gaze input with a configuration-driven backend for swappable spatial filtering, neural speech separation (FasNet/TasNet), and ASR

ReSQL - SQL Service Layer for Apache ResilientDB (Open Source contribution)

- Contributed to **ReSQL**, a SQL service layer that embeds DuckDB as an in-process analytical backend on top of ResilientDB's **distributed BFT ledger**, enabling relational schemas and SQL semantics over its key-value execution model
- Implemented **DuckDB storage integration**, proto definitions, and Bazel build wiring; refactored runtime config so SQL execution can be selectively enabled via flags without affecting KV apps or core consensus

Drug Supply Chain Tracking on a Decentralized Network

- Built a React/Node.js web3 app with **Solidity smart contracts** and MongoDB to secure transactions in a pharmaceutical drug supply chain; [published a paper](#) in the **Indonesian Journal of Electrical Engineering and Computer Science (IJECS)**, Jan 2024, Vol. 33

TECHNICAL SKILLS

Programming Languages: Python, JavaScript / TypeScript, Java, Solidity

Frameworks: Django, Node.js, Electron, Express, ReactJS, LangChain, LangGraph

Database Technologies: MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, ChromaDB

Cloud & DevOps: AWS (S3, Lambda), Docker, Git/GitHub